Ans 1. Intelligent Agents in a Smart Home Automation System.

Introduction:

An intelligent agent is a system that observes its environment through sensors and takes suitable actions to achieve specific objectives. In a smart home, intelligent agents control lighting, temperature and security using motion sensors, temperature sensors, cameras and time based information. They can also learn user preferences to improve comfort and energy efficiency.

Types of Intelligent Agents:

1) Simple Reflex Agent
→ Works based on the current percept using condition - action rules.
→ Does not remember past events or learn user preferences.
→ Example: Switches ON lights when motion is detected.

2) Model - Based Agent.
→ Maintains an internal state using current and past information.
→ Example: Tracks room occupancy to control lighting.

3) Goal - Based Agent.
→ Takes actions to achieve specific goals.
→ Example: Adjusts room temperature to the desired level.

4) Utility Based Agent
→ Selects actions that provide the best overall benefit.
→ Example: Balances comfort, electricity usage and security.

Comparison:

Agent	Application
1) Simple reflex	Automatic lighting
2) Model Based	Occupancy Tracking
3) Goal Based	Temperature control
4) Utility Based	Energy optimization

Best Approach:

A hybrid approach combining all four agents is most effective.

A hybrid intelligent agent provides comfort, energy efficiency, personalized control and better security by combining immediate responses with intelligent decision making.

Ans 2. Uninformed Search strategies.

1) ~~Or~~ Search strategies
   → Expands the lowest-cost path first.
   → Advantage: Finds the least-cost solution
   → Disadvantage: Requires more time and memory.
   → Time & Space: Exponential in the worst case.

2) Interactive Deepening Search (IDS).
   → Increases the depth limit repeatedly.
   → Advantage: Uses less memory.
   → Disadvantage: Repeats searches.

3) Types of Intelligent Agents
   → simple reflex: Reacts to current sensor inputs.
   → Model based: Remembers explored paths & obstacles.
   → Goal based: Plans to reach the text.
   → Utility based: Choose paths based on cost and safety.

4) Best approach:
   A model-based, goal-based agent with UCS is best for finding the least-cost path while remembering obstacles. IDS is preferable when memory is limited and all moves have equal cost.

5) Conclusion:
   Combining agent memory, goal-oriented planning, and a suitable search strategy helps the robot navigate the unknown maze efficiently.